SESCS — Master Index

Self-Evolving Systems Control Standard

Governance & Enforcement · Autonomous Systems Control

System Overview

The **Self-Evolving Systems Control Standard (SESCS)** defines a complete control architecture for systems capable of modifying their own behavior.

It establishes a closed-loop governance system composed of five interconnected modules:

- control
- traceability
- validation
- responsibility
- recovery

Together, these modules form a **continuous control cycle.**

Core System Logic

SESCS operates as a closed governance loop:

Control → defines what can change
Logging → records what changed
Validation → determines if change is acceptable
Responsibility → assigns accountability
Recovery → restores system integrity

This loop is continuous and non-bypassable.

Module Structure

1. MCM — Mutation Control Module

Defines boundaries and permissions for system mutation.

→ No mutation outside defined zones

2. MLM — Mutation Logging Module

Ensures full traceability of all system changes.

→ No log → no trust

3. MVM — Mutation Validation Module

Validates mutations before execution.

→ No validation → no execution

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4. RAM — Responsibility Assignment Module

Assigns accountability to every action.

→ No responsibility → no system

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5. RIM — Rollback & Integrity Module

Ensures recovery and system integrity.

→ No rollback → no recovery

System Condition

A system is considered:

CES (Controlled Evolution System) → all modules active
USS (Uncontrolled System State) → any module missing

Structural Rule

All five modules must operate together.
Partial implementation is not valid.

System Flow

The system operates in continuous cycles:

Mutation → Control → Logging → Validation →
Responsibility → Recovery → System Integrity

This ensures that:

- no change is uncontrolled
- no action is untracked
- no decision is unvalidated
- no outcome is unassigned
- no failure is irreversible

Compatibility

SESCS integrates with:

- MTVF — validation layer

- TSA — trust classification
- EVIP — ethical governance
- ArtData — validated data layer
- ORGS — system compatibility

Core Insight

Self-evolving systems are not dangerous because they evolve.
They are dangerous when they evolve without control.

Closing Statement

SESCS defines the minimum structure required to safely operate self-evolving systems.

It transforms system evolution from:

uncontrolled change → governed process

Modules

→ Module 1 — MCM: Mutation Control Module

→ Module 2 — MLM: Mutation Logging Module

→ Module 3 — MVM: Mutation Validation Module

→ Module 4 — RAM: Responsibility Assignment Module

→ Module 5 — RIM: Rollback & Integrity Module

Related Documents

→ SESCS Standard

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